

Sanghyun Hahn

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EDUCATION

Seoul National University (SNU), College of Engineering , Seoul, Korea	Feb. 2026
B.S. in Aerospace Engineering	GPA: 4.14/4.30 (3.95/4.00)
Graduation with Summa Cum Laude, Rank 1/36	
Seoul Science High School , Seoul, Korea	Feb. 2020
Specialized in Mathematics and Physics	GPA: 4.16/4.30

RESEARCH EXPERIENCE

SNU Machine Perception and Reasoning Lab , Seoul, Korea	
<i>Undergraduate Researcher</i> (Advisor: Prof. Jonghyun Choi)	Mar. 2025 – Present
- Applied action chunking to dexterous grasping by modifying PPO. Outperformed all PPO-based methods in success rate and training time. Presented at IEEE Humanoids 2025 Workshop; preparing for NeurIPS 2026. - Reformulated manipulation tasks into a 3D matching problem. Utilized Gaussian Splatting as SE(3) equivariant features for one-shot imitation learning across novel object poses and instances.	
SNU Lab for Autonomous Robotics Research , Seoul, Korea	
<i>Undergraduate Research Intern</i> (Advisor: Prof. Hyoun Jin Kim)	Sep. 2024 – Feb. 2025
Improved Gaussian Splatting by clustering-based seeding in under-reconstructed regions and integrating reconstruction error for opacity initialization. Awarded a \$700 scholarship from SNU AeroDrone.	
SNU Robust Perception and Mobile Robotics Lab , Seoul, Korea	
<i>Undergraduate Research Intern</i> (Advisor: Prof. Ayoun Kim)	Jul. 2023 – Aug. 2024
- Proposed a target-based accuracy evaluation metric for LiDAR-Inertial SLAM. Raised \$3,500 in funding through the undergraduate research program at SNU. Presented at ICCAS 2024. - Developed a LiDAR–Thermal camera system on a UGV; led design, manufacturing, calibration, and control.	

PUBLICATIONS

- Hahn, S., & Choi, J.** Action Chunking Proximal Policy Optimization for Universal Dexterous Grasping. *IEEE Humanoids 2025 Workshop on Dexterous Human Manipulation*, 2025.
- Hahn, S., Oh, S., Jung, M., Kim, A., & Jung, S.** Quantitative 3D Map Accuracy Evaluation Hardware and Algorithm for LiDAR(-Inertial) SLAM. *IEEE ICCAS*, 2024.

TEACHING

Seminar Organizer – Reinforcement Learning for Robotics SNU	Fall 2025
Founded a peer-learning seminar (1 Credit); designed syllabus, selected papers, hosted weekly sessions.	
SPLIT Tutor in Physics SNU	Jan. 2022 – Feb. 2022
Delivered weekly lectures and review sessions for incoming freshmen on general physics.	

HONORS & AWARDS

Student-Directed Education Undergraduate Research Program SNU	Fall 2024
National Scholarship for Science and Engineering Korea Student Aid Foundation	2020 – 2024
Full tuition scholarship for eight semesters.	

ADDITIONAL EXPERIENCE & LEADERSHIP

Reviewer IEEE RA-L	2025
Translator and Tutor EL. Brown Academy	Feb. 2020 – Mar. 2023
- Translated JEE Advanced Physics materials and developed SNU entrance exam practice tests. - Taught mathematics to high school student groups for school exams and the SNU entrance exam. - Mar. 2022 – Nov. 2022: On leave for mandatory military service	
Supply Specialist Republic of Korea Army	May 2022 – Oct. 2022
Mandatory military service, medically discharged.	
Private Tutor Self-Employed	Feb. 2020 – Apr. 2022
Mentored a high school student in mathematics, leading to acceptance into SNU.	
Powertrain Team Leader SNU Baja Student Team	Dec. 2020 – Nov. 2021
Led powertrain and chassis design. Supervised mechanical fabrication/testing teams.	

SELECTED COURSES

Engineering: Control Theory, Structural Analysis, Sensor Systems, Robot Vision

Mathematics: Linear Algebra, Stochastic Processes, Mathematics of Neural Networks, Mathematics of RL & LLMs

AI/CS: Machine Learning, Deep Learning, 3D Computer Vision, Algorithms, Topics in Computer & VLSI

SELECTED COURSE PROJECTS

Forward Facing 3D Gaussian Splatting as Markov Chain Monte Carlo , Deep Learning	Fall 2024
Removed floating artifacts in forward-facing scenarios for 3DGS-MCMC. Selected for oral presentation.	
OceanGate Titan Structural Analysis , Mechanics of Aerospace Structures (Structural Analysis)	Fall 2023
Simulated the effects of thermal shock and vertical impact on ANSYS. Selected for oral presentation.	

SKILLS & INTERESTS

Languages: Korean (Native), English (Fluent)

Programming & Frameworks: Python, C, MATLAB, PyTorch, JAX/Flax, ROS

Simulation: Isaac Gym, Isaac Sim, Gazebo, ANSYS

Hardware: Welding, Machining, Circuits, 3D Printing, Sensors (LiDAR, RGB-D/Thermal Camera)

Research Interests: Humanoids, Dexterous Manipulation, Reinforcement Learning, Robot Learning